
Fingerprint Quality Assessment Report

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Page 1 - Report Questions

1. What threshold did you set for blur, and how did you decide it?

I used a main Laplacian threshold of 300, a severe threshold of 70, and a strip-spread threshold of 80. I calibrated them by comparing five good and five blurry phone images until all blurry samples failed without rejecting acceptable captures.

2. Which metric was hardest to implement correctly, and what went wrong first?

Blur was hardest because whole-image Laplacian variance measured sharp cables, screens, and walls instead of the blurred finger. I solved this with three overlapping central strips plus median, severe-blur, and score-spread rules.

3. What is NFIQ2, and why is it unreliable for phone-camera images?

NFIQ2 is NIST Fingerprint Image Quality 2, designed mainly for contact-scanner fingerprint images. Phone images vary in perspective, scale, lighting, focus, and background, so contact-scanner quality features are not directly calibrated for them.

4. Name three additional quality checks for a real deployment.

I would add dirt or wound detection, multiple-finger or object-in-frame detection, and uneven shadow detection. These checks would reject contaminated fingertips, ambiguous captures, and images where part of the ridge area is hidden by non-uniform lighting.

5. If a rural agricultural worker's fingerprints are naturally worn and give consistently poor ridge clarity scores, what should the system do differently for them?

The system should avoid repeated automatic rejection and use adaptive thresholds based on repeated captures. It should also allow another finger, assisted recapture, or an alternative verification method for fair access.

Module Findings and Fixes

Summary of what happened in each implementation module and how it was resolved.

1. Setup and preprocessing	Added image validation, grayscale conversion, and resizing to a maximum dimension of 1024 pixels. This kept input handling consistent and reduced processing time.
2. Blur detection	Whole-frame and mask-based Laplacian scores were dominated by sharp backgrounds. Three central strips with median, severe, and spread rules correctly separated all five good and five blurry samples.
3. Brightness	Global brightness was misleading when the background differed from the finger. Measuring the mean inside the detected finger ROI produced reliable dark and bright decisions using the 80-210 range.
4. Glare detection	A full-image bright-pixel fraction missed small fingertip reflections. Combining global glare, ROI glare, low saturation, and the largest connected hotspot detected all five glare samples.
5. ROI completeness	Otsu thresholding initially selected background regions in difficult scenes. Gaussian smoothing, morphology, contour selection, and a 15% coverage threshold provided a practical finger-size check.
6. Ridge clarity	Gabor responses were affected by mask boundaries and weak contrast. Fingertip extraction, a central ridge-core mask, CLAHE, erosion, and six filter orientations stabilized the ridge score.
7. Composite scoring and guidance	Normalized metrics were combined with weights into a 0-100 score, while every metric remained mandatory. Failed captures are capped at 59.99 and receive direct recapture guidance.
8. Evaluation and calibration	The test script evaluated five good, five blurry, five dark, and five glare images and saved detailed results to CSV. Final calibration produced 20/20 correct results on this dataset.
9. Performance	Resizing images, reusing the finger mask, and limiting Gabor analysis to the fingertip core reduced average processing time to about 143 ms, below the 300 ms requirement.
10. Streamlit and testing	The UI added image upload, adjustable thresholds, metric cards, timing, and guidance while deleting temporary files. Pytest completed successfully with one passing end-to-end test.